Prem P. Singh, Ph.D.

Scientist & Computational Biologist | Translating Noisy Ag-Bio Data into Validated Assays & Actionable Insights
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Professional Summary

Scientist & Computational Biologist with 6+ years delivering **'bench-to-pipeline'** molecular diagnostics and high-throughput genomic pipelines (dPCR/qPCR, NGS, >1.3B reads). LBNL-trained in Scientific Leadership & Management. Transitioning to industry to formalize academic R&D into SOP/GxP-compliant, commercial-grade pipelines.

Professional Experience

Postdoctoral Scholar 2023 – Present
University of California, Davis *Davis, CA*

- Executed CDFA/USDA surveillance across 10+ commercial vineyards; **enabled pathogen detection 2–3 months before visual symptoms**.
- Managed a 300+ sample RNA-seq study (>1.3B reads); automated 96-well extraction on MagMAX Express-96, achieving **5× throughput increase**.
- Developed and validated a dPCR diagnostic assay with TaqMan internal controls; **eliminated false negatives across 100% of field isolates**.
- Built an ML classifier (XGBoost) identifying pre-symptomatic biomarkers with **2.29× signal enrichment** near trait loci (GitHub).
- Engineered the field's first longitudinal mixed-effects epidemiological model, decoupling heat-stress confounders from viral kinetics across 4 growth stages.
- Built a developmental clock on healthy vines (61/63 correct, whole-vine withheld); validated with **1,999 nested bootstrap draws and 4,999 vine-label permutations**, calibrating genome-wide interactions against 200 permutation refits.
- Coordinated 3 cross-functional teams (USDA-ARS, UC Davis, wineries); **cut sample-to-report turnaround by 50%** via standardized SOPs.

MOVE Fellow (AI Trainer) Sep 2025 – Present
Handshake AI *Remote*

- Audited **600+ scientific prompt-response pairs** across frontier LLMs, verifying factual accuracy in plant genetics, molecular biology, and biochemistry.
- Authored **150+ gold-standard SFT reference datasets** and established error taxonomies to mitigate hallucinations and fabricated citations for RLHF alignment.

Doctoral Researcher 2017 – 2023
Banaras Hindu University *Varanasi, India*

- Engineered a synergistic botanical formulation via DoE ($p = 0.0083$); achieved **85% fungal reduction and 100% aflatoxin B1 elimination** in 6-month storage trials.
- Synthesized a chitosan-cinnamic acid nanogel (20–70 nm) with **60-day sustained release** and 62% thermal retention up to 364 °C.
- Ran **2.0 μs all-atom MD simulations** (Amber16, MM-PBSA) across 6 enzyme targets; **cut mechanistic validation from 3 months to 3 weeks**.
- Executed the first whole-transcriptome study of the mold ($n = 12$; paired-end RNA-seq; FDR $q < 0.05$); validated by RT-qPCR.
- Generated the first *de novo* genome assembly for a regional toxigenic strain (**99.5% coverage**; GenBank MZ497370); annotated 80 biosynthetic gene clusters.
- Validated **100% aflatoxin decontamination** (HPLC-PDA) in 6-month in situ trials while fully preserving grain nutritional quality.
- Built a reproducible HPC pipeline (Snakemake, SLURM) adopted by 2–3 partner labs; authored 6 first-author papers and mentored 4 junior researchers.

Technical Skills

Programming & Pipelines	Python (Scikit-learn, Pandas), R/Bioconductor, Bash, Snakemake, SLURM, Linux/HPC, Singularity, Git/GitHub
Genomics & Transcriptomics	Bulk RNA-seq, TAG-seq, Differential gene expression, Gene set enrichment, Pathway mapping, Multi-omics factor analysis, UMI quantification
Laboratory & Analytical Techniques	RT-qPCR, Digital PCR (dPCR/ddPCR), Primer/Probe design, LOD/LOQ validation, NGS library prep, MagMAX Express-96 automation, SOP authoring, HPLC-PDA, GC-MS, LC-MS/MS, FT-IR, UV-Vis, TGA, Nanoencapsulation, Controlled-release testing
Statistics, ML & Experimental Design	Design of Experiments (DoE), Response-surface modeling, Bayesian modeling, Biomarker discovery, XGBoost, Mediation analysis, Mixed-effects models
Domain Knowledge	Plant virology, Fungal pathogenesis, Mycotoxin biology, Infection bioassays, Field disease surveillance, Crop protection, PGPR/Plant-microbe interactions, Biopesticide formulation

Selected Projects

Microsecond Molecular Dynamics & Target Discovery | *Amber16, MM-PBSA, AutoDock, Python, HPC* 2020 – 2022

- Executed 2.0 μs all-atom MD across 6 enzyme complexes; cut target validation from 3 months to 3 weeks; resolved multi-target inhibition of aflatoxin biosynthetic enzymes.

Synergistic Botanical Formulation & DoE Platform | *JMP, SPSS, GC-MS, HPLC-PDA* 2019 – 2023

- Optimized a ternary botanical blend via augmented simplex-centroid DoE ($p = 0.0083$); achieved 85% fungal reduction and 100% aflatoxin B1 elimination in 6-month sorghum storage trials.

Chitosan Nanopolymer Drug Delivery System | *SEM, TGA, FT-IR, XRD, UV-Vis* 2019 – 2022

- Synthesized a crosslinked chitosan-cinnamic acid nanogel (20–70 nm); achieved 62% thermal retention up to 364°C and sustained biphasic release extending shelf-life to 60 days.

Rhizosphere Microbiome & PGPR Bio-Inoculants | *16S rRNA, CAS Agar, Salkowski Assay* 2015 – 2017

- Isolated and characterized plant-growth-promoting rhizobacteria from medicinal crop rhizospheres; profiled phosphate solubilization, IAA production, and siderophore synthesis. Contributed to $\sim$10 papers and 10 book chapters.

Education

Ph.D., Plant Pathology & Computational Biology – Banaras Hindu University 2023
Masters, Botany/Plant Pathology (*Gold Medalist*) – Banaras Hindu University 2017
Bachelors, Plant Sciences – Banaras Hindu University 2015

Conferences & Professional Development

Scientific Leadership & Management Skills Program: Lawrence Berkeley National Laboratory (LBNL), 2025
International Workshop on Genome Informatics: EMBL-EBI Ensembl Outreach, one-month residency, Hinxton, UK, 2023
7th International Conference on Phytopathology: ICAR-IARI, New Delhi (*Best Poster Award*), 2020
International Conference on Microbiology (ICMBB2021): Asian PGPR Society/American Society for Microbiology (ASM), 2021

Selected Publications

- **Singh, P.P.**, Scully, H., Reddy, K., Boghozian, A.M., Medina-Plaza, C., Oberholster, A., and Sudarshana, M.R. (2026). Temporal dynamics of Grapevine red blotch virus accumulation in grapevine leaves are influenced by fruit maturity stages. *Archives of Virology*.
- **Singh, P.P.**, Singh, R., Verma, P.K., and Prakash, B. (2026). Systematic investigation of aflatoxigenic *Aspergillus flavus* inhibition: integrating essential-oils-based formulation with mathematical modeling and transcriptomic analysis. *International Journal of Food Microbiology*, 111632.
- **Singh, P.P.**, Jaiswal, A.K., Kumar, A., Gupta, V., and Prakash, B. (2021). Untangling the multi-regime molecular mechanism of verbenol-chemotype *Zingiber officinale* essential oil against *Aspergillus flavus* and aflatoxin B1. *Scientific Reports*, 11, 6832.